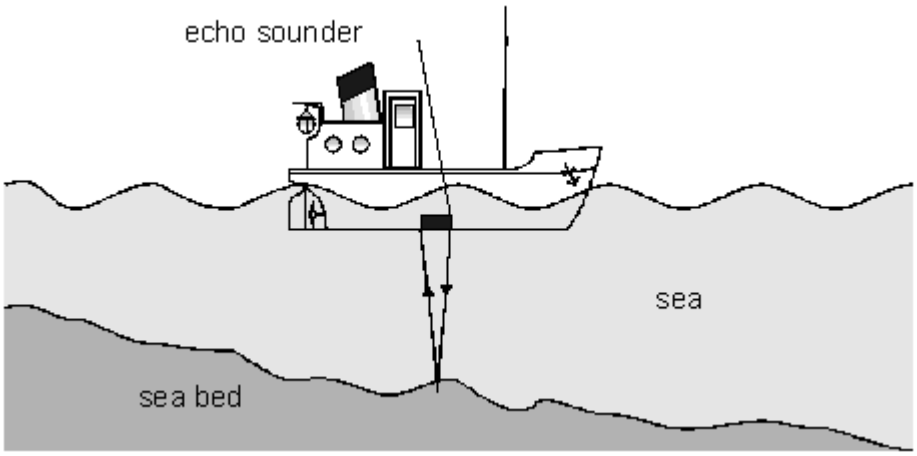
Q1.

The diagram shows a boat using an echo sounder. It sends a pulse of sound waves which is reflected from the sea bottom. The reflected sound waves are detected by a sensitive microphone.



The time between sending and receiving the pulse is 0.005 s. The device calculates the depth of the sea, using the speed of sound in sea water, which is 1500 m/s.

(a) Calculate the depth of the sea. Show your working.

.....

.....

.....

2 marks

(b) The boat moves into very deep water. Explain why the reflected pulse is too weak to be detected.

.....

.....

1 mark

(c) The boat's 'echo sounder' could **not** be used in an aeroplane to measure its height above the ground unless it had been modified.

Explain why the device will **not** give correct heights above the ground.

.....

.....

1 mark

Maximum 4 marks

Q1.

(a) distance travelled by pulse = 7.5m

accept 'time to sea bed = 0.0025s'

1

depth of sea = 3.75m

the unit is required for the mark consequential marking applies
accept half the distance travelled by the pulse accept the correct
answer with no working for both marks

1

(b) any **one** from

- the energy spreads out

*accept 'it **or** the signal spreads out'*

- the energy is absorbed by the water

*accept 'it **or** the signal is absorbed by the water'*
accept 'it is drowned out by noise'
*accept 'the echo sounder **or** the signal made by the echo sounder is **not** powerful enough'*
*do **not** accept 'the signal is not powerful enough'*

1

(c) because the speed of sound in air is different from that in water

accept 'the speed of sound in air is less'
accept 'the speed of sound in air is 330m/s'

1

[4]